

Executive Summary



Sponge City for Everyone

Jakarta's flood risk is rising—driven largely by land subsidence and land-use change. A practical response is the **Sponge City** approach: manage rain where it falls so it soaks in, is stored, or reused instead of rushing into drains and causing floods. One tangible measure is porous (permeable) paving, but today its adoption is still concentrated among **higher-income users**, limiting citywide impact.

In this project, **The Yellow Sheet** team formulated four scenarios which we tested using the runoff coefficient (C)—a 0–1 number showing the fraction of rainfall that becomes runoff (lower is better):

- 1 Current condition
- 2 Early adopters install porous paving block
- 3 Government targets high-risk areas
- 4 Citywide coverage on suitable surfaces

City-wide C is 0.733 today. **Early-adopter** action barely changes it (0.732). **Targeting high-risk areas** performs better (0.726). The strongest result comes from a full implementation, which lowers C to **0.696**—about a **5% reduction**—with decreases across all municipalities.

Jakarta can move from **concept to practice** by scaling porous paving blocks and other complementary measures. Our scenarios show that **depending to early adopter only changes little**, while **prioritizing high-risk areas delivers bigger results**, around 5% drop in city-wide run-off coefficient.

One pilot project in five sub-districts in high-risk area shows the **potential runoff reduction of 1 million m³ per year**. This can be the start for DKI Jakarta to widely implement by replicating pilot projects and tracking results with a simple SOP—with government, communities, and developers acting together.

Let's build a Sponge City for Everyone.